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<!DOCTYPE html>
<html lang="es"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>Dos ciudades, 1982</title>
<style>
html,body{margin:0;background:#0b0e1a;color:#ffe9a8;font:12px monospace;height:100%}
#wrap{overflow-x:auto;overflow-y:hidden;width:100vw;height:100vh;display:flex;align-items:center}
#stage{position:relative;margin:auto;flex:none}
canvas{display:block;image-rendering:pixelated;image-rendering:crisp-edges}
#ov{position:absolute;inset:0;pointer-events:none}
.lb{position:absolute;transform:translate(-50%,-100%);background:#0b0e1add;border:1px solid #ffe9a8;padding:1px 3px;font-size:10px;white-space:nowrap}
#ui{position:fixed;right:6px;top:6px;display:flex;gap:4px;z-index:3}
button{font:11px monospace;background:#1c2340;color:#ffe9a8;border:1px solid #ffe9a8;padding:4px 6px}
#rot{position:fixed;left:0;right:0;bottom:6px;text-align:center;font-size:11px;display:none}
@media (orientation:portrait){#rot{display:block}}
</style></head><body>
<div id="wrap"><div id="stage"><canvas id="c"></canvas><div id="ov"></div></div></div>
<div id="ui"><button id="bg">Guía</button><button id="bs">Sonido</button></div>
<div id="rot">Girá el teléfono o deslízalo de costado ↔</div>
<script>
"use strict";
// ===== PARÁMETROS =====
const Q=new URLSearchParams(location.search);
const W=480,H=270;
const LOOP=96; // duración del loop (s)
let SPEED+=Q.get('speed')||1; // velocidad (?speed=2)
const START+=Q.get('t')||0; // segundo inicial (?t=80)
// ===== PALETA FIJA (32 colores) =====
const
PAL=["#0b0e1a","#1c2340","#3a4a7a","#6f8fc9","#a9c6ea","#f2c28a","#e07a4f","#ffe9a8",
"#2a2a33","#4a4a55","#7a7a86","#b5b5bd","#e8e4d8","#6b3e2e","#a5563a","#c93434",
"#f06a8a","#3fd0e0","#f5d547","#3c7a3c","#7fc05a","#2b5f8a","#4d8fbf","#8a6a3a",
"#d9b45a","#e8b8a0","#8a5a3c","#1e3a5f","#5a4a7a","#ff9f40","#ffffff","#000000"];
// equivalente nocturno de cada color (siempre dentro de la paleta)
const
DK={2:1,3:2,4:2,5:14,6:13,7:29,9:8,10:9,11:9,12:10,13:8,14:13,15:13,16:28,17:21,18:29,19:8,
20:19,21:27,22:21,23:8,24:23,25:14,26:13,27:1,28:1,29:14,30:11};
const cv=document.getElementById('c'),dc=cv.getContext('2d');
const buf=document.createElement('canvas');buf.width=W;buf.height=H;
const g=buf.getContext('2d');
const B4=[0,8,2,10,12,4,14,6,3,11,1,9,15,7,13,5];
const bay=(x,y)=>(B4[(y&3)*4+(x&3)]+.5)/16;
function hash(a,b){let
h=(a*374761393+b*668265263)|0;h=Math.imul(h^(h>>>13),1274126177);return
((h^(h>>>16))>>>0)/4294967296}

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const mod=(a,n)=>((a%n)+n)%n,lerp=(a,b,f)=>a+(b-a)*f;
let T=0,P=0,N=0;
// k vueltas enteras por loop => periodicidad exacta
const cyc=(k,span,off)=>mod((off||0)+k*span*T/LOOP,span);
const osc=(k,ph)=>Math.sin(2*Math.PI*k*T/LOOP+(ph||0));
const tick=r=>Math.floor(mod(T*r,LOOP*r)); // r entero => periódico
// progreso 0..1 de un micro-evento que empieza en s y dura d (o -1)
function uOf(s,d){const l=mod(T-s,LOOP);return l<d?!l/d:-1}
function walkIn(xt,u,from,to){return u<.15?lerp(from,xt,u/.15):u>.85?lerp(xt,to,(u-.85)/.15):xt}
function R(c,x,y,w,h){ // color que se oscurece de noche
  if(N>0&&DK[c]!==undefined&&bay(x|0,y|0)<N*.9)c=DK[c];
  g.fillStyle=PAL[c];g.fillRect(x|0,y|0,w|0,h|0)}
function E(c,x,y,w,h){g.fillStyle=PAL[c];g.fillRect(x|0,y|0,w|0,h|0)} // emisor
// texto pixelado cacheado y umbralizado a un color de paleta
const TC={};
function txt(s,x,y,c){const k=s+'|'+c;let o=TC[k];
  if(!o){const m=document.createElement('canvas'),mx=m.getContext('2d');mx.font='bold 6px monospace';
    const w=Math.ceil(mx.measureText(s).width)+2;m.width=w;m.height=8;
    mx.font='bold 6px monospace';mx.textBaseline='top';mx.fillStyle='#fff';mx.fillText(s,1,1);
    const
d=mx.getImageData(0,0,w,8),a=d.data,hx=PAL[c],r=parseInt(hx.substr(1,2),16),gg=parseInt(
hx.substr(3,2),16),b=parseInt(hx.substr(5,2),16);
    for(let i=0;i<a.length;i+=4){const on=a[i+3]>110;a[i]=r;a[i+1]=gg;a[i+2]=b;a[i+3]=on?255:0}
    mx.putImageData(d,0,0);o=TC[k]=m}
  g.drawImage(o,x|0,y|0);return o.width}
// ===== CICLO DÍA/NOCHE =====
function nightOf(p){if(p<.68)return p<.08?1-p/.08:0;if(p<.78)return (p-.68)/.1;return 1}
const SKY=[[0,1,2],[.08,6,5],[.14,3,4],[.48,3,4],[.6,5,6],[.7,6,28],[.78,1,2],[.92,0,1],[1,1,2]];
function drawSky(){let i=0;while(SKY[i+1][0]<=P)i++;const
a=SKY[i],b=SKY[i+1],f=(P-a[0])/(b[0]-a[0]);
  for(let y=0;y<200;y+=2){const v=y/200;for(let x=0;x<W;x+=2){const
k=bay(x>>1,y>>1)<f?b:a;
    E(bay((x>>1)+2,(y>>1)+1)<v?k[2]:k[1],x,y,2,2)}}}
function drawCelestial(){
  if(N>0)for(let
i=0;i<60;i++){if(hash(i,3)<N&&(tick(2)+i)%9)E(hash(i,4)>.8?7:12,hash(i,1)*W,hash(i,2)*110,1,
1)}
  if(P<.7){const
u=P/.7,x=20+u*440,y=150-Math.sin(u*Math.PI)*120;E(7,x-4,y-4,9,9);E(18,x-3,y-3,7,7)}
  else if(P>.72&&P<.98){const
u=(P-.72)/.26,x=40+u*400,y=150-Math.sin(u*Math.PI)*110;E(12,x-3,y-3,7,7);E(11,x,y-2,3,3)}
  const us=uOf(77,11);if(us>=0){const
x=us*W,y=30+us*20;E(30,x,y,1,1);if(tick(4)%2)E(15,x-2,y,1,1)} // satélite compartido
}
function birds(){for(let i=0;i<7;i++){const
x=cyc(2,W+80,i*9)-40,y=60+osc(12,i)*6+(i%3)*5,w=(tick(6)+i)&1,c=N>.5?9:8;
  E(c,x,y,1,1);E(c,x-1,y-w,1,1);E(c,x+1,y-w,1,1)}}

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function drawRiver(){R(27,0,240,W,4);R(21,0,244,W,26);
  for(let i=0;i<40;i++)E(22,cyc(1+(i%3),W+20,hash(i,5)*W)-10,245+(hash(i,6)*23|0),4,1);
  if(N>.4)for(let i=0;i<30;i++){const
x=hash(i,8)*W|0;if((tick(4)+i)%3)E(x<240?17:29,x,244+(hash(i,9)*24|0),2,1)}
  const
bx=cyc(1,W+60,0)-40;E(8,bx,252,34,4);E(15,bx+22,247,8,5);E(12,bx+2,249,18,3);if(N>.4)E(
7,bx+24,248,2,1)}
// ===== PIEZAS =====
function winGrid(x,y,w,h,dx,dy,base,seed){for(let j=y+2;j<y+h-3;j+=dy)for(let
i=x+2;i<x+w-2;i+=dx){
  if(hash(seed+i,j)<N*.75)E(7,i,j,1,2);else R(base,i,j,1,2)}}
function P8(x,y,top,bot,skin,f,o){o=o||{};x|=0;y|=0;const st=f&1;
  if(o.sign){R(9,x+1,y-16,1,6);R(12,x-3,y-21,10,5);R(15,x-2,y-19,8,1)}
  R(o.hair!=null?o.hair:8,x+1,y-10,2,1);R(skin,x+1,y-9,2,2);R(top,x,y-7,4,5);
  if(o.scarf)R(15,x+1,y-7,2,1);
  R(bot,x+st,y-2,1,2);R(bot,x+3-st,y-2,1,2);
  if(o.bag!=null)R(o.bag,x+4,y-4,2,2);
  if(o.arm)R(skin,x+4,y-11,1,4);
  if(o.hat!=null)R(o.hat,x,y-11,4,2);
  if(o.phones){R(15,x,y-10,1,2);R(15,x+3,y-10,1,2)}}
function
P5(x,y,top,f,scarf){x|=0;R(25,x,y-6,2,2);R(top,x,y-4,3,3);if(scarf)R(15,x,y-4,3,1);R(8,x+(f&1),y-
1,1,1);R(8,x+2-(f&1),y-1,1,1)}
function
car(x,y,c,dir,taxi){x|=0;R(c,x,y-5,14,4);R(c,x+3,y-8,8,3);R(4,x+4,y-7,2,2);R(4,x+8,y-7,2,2);
  R(8,x+2,y-1,3,2);R(8,x+10,y-1,3,2);if(taxi)E(18,x+6,y-9,2,1);
  if(N>.35){E(7,dir>0?x+13:x,y-4,1,1);E(15,dir>0?x:x+13,y-4,1,1)}}
function lamp(x,warm){R(9,x,176,1,31);R(9,x-2,176,3,1);
  if(N>.35){const c=warm?29:7;E(c,x-2,177,2,1);for(let y=180;y<207;y+=2){const
s=(y-180)>>2;for(let i=-2-s;i<=s;i+=2)E(c,x+i,y,1,1)}}}
function crane(mx,base,top,j0,j1,hx,yLow,yTop,u,lc){R(29,mx,top,2,base-top);for(let
y=top+2;y<base;y+=4)R(8,mx,y,2,1);
  R(29,j0,top,j1-j0,2);R(9,j0,top+2,4,3);let y=yLow,ld=false;
  if(u>=0){if(u<.45){y=lerp(yLow,yTop,u/.45);ld=true}else if(u<.6){y=yTop;ld=u<.55}else
y=lerp(yTop,yLow,(u-.6)/.4)}
  R(9,hx,top+2,1,y-top-2);R(8,hx-1,y,3,2);if(ld)R(lc,hx-5,y+2,10,3)}
function puffs(cx,top,n,dark){for(let i=0;i<n;i++){const
a=mod(tick(6)+i*8,48),y=top-a*.8;if(y<0)continue;
  R(dark?(i%2?8:9):(i%2?10:11),cx+2+a*.7+osc(3,i)*2,y,2+a/10,2+a/12)}}
const TK="ACME +1/4 ZENO -3/8 ORBX +1/2 KAPO -1/8 ";
// ===== PEATONES (8 por lado): [vueltas, desfase, dir, y, top, bot, piel, extras] =====
const
CL=[[3,0,1,206,27,8,25,{bag:13}],[4,40,-1,206,9,8,26,{bag:13}],[2,90,1,206,15,27,25,{hair:13}
],[3,130,-1,206,29,21,26,{hat:18}],
[2,20,1,234,12,21,26,{hair:31}],[4,70,-1,234,27,8,25,{bag:13}],[3,120,1,234,14,23,25,{}],[2,18
0,-1,234,20,21,26,{}]];

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const
CR=[[3,10,1,206,21,21,25,{hat:9}],[2,50,-1,206,23,8,25,{hat:16,bag:23}],[4,100,1,206,15,9,25,
,{hat:15}],[3,140,-1,206,21,27,26,{}],
[2,30,1,234,20,9,25,{hat:16}],[4,80,-1,234,9,8,26,{hair:11}],[3,130,1,234,27,27,25,{bag:13}],[2,
,200,-1,234,14,23,25,{}]];
function crowd(L,x0){L.forEach((p,i)=>{const
c=cyc(p[0],260,p[1]);P8(p[2]>0?x0-10+c:x0+250-c,p[3],p[4],p[5],p[6],tick(6)+i,p[7]))}
// ===== CIUDAD CAPITALISTA (x 0..239) =====
function drawLeft(){let u;
  for(let i=0;i<13;i++){const h=50+hash(i,7)*80|0;R(2,i*19-5,200-h,16,h)}
  R(27,4,50,34,150);winGrid(4,52,34,146,3,4,22,11);R(9,20,38,2,12); // torre de vidrio
  R(13,40,105,26,95);winGrid(40,105,26,78,4,5,9,23);for(let
x=40;x<66;x+=6)R(9,x,88,1,17);R(9,40,88,26,1); // obra de ladrillo
  R(12,42,181,22,5);txt("CLOSING",42,180,15);R(0,42,186,22,14);txt("SALE",45,188,18);
  u=uOf(52,14);const sh=u<0?(mod(T-52,LOOP)<40?1:0):Math.min(1,u*2); // persiana
  baja y se levanta al amanecer
  for(let y=186;y<186+14*sh;y+=2){R(10,42,y,22,1);R(9,42,y+1,22,1)}
  R(11,68,118,40,82);R(12,66,113,44,5);txt("CIVIC BANK",67,112,27);R(0,68,120,40,6); //
  banco con cotizaciones
  g.save();g.beginPath();g.rect(68,120,40,6);g.clip();const tw=txt(TK,-999,0,18);const
  ox=cyc(6,tw,0);txt(TK,68-ox,119,18);txt(TK,68-ox+tw,119,18);g.restore();
  for(let i=0;i<5;i++)R(12,71+i*8,128,3,62);R(10,64,195,48,5);

R(10,110,62,30,138);R(10,114,44,22,18);R(11,119,30,12,14);R(11,124,12,2,18);winGrid(110,
64,30,134,3,4,4,31); // escalonado
  R(14,142,130,44,70);txt("MERCER&DAYE",142,130,12);winGrid(142,138,44,38,4,5,5,41);
  // tienda departamental
  for(let i=0;i<11;i++)R(i%2?12:15,142+i*4,176,4,3);R(1,144,182,40,16);
  for(let i=0;i<3;i++){E(9,147+i*12,185,9,7);E([17,12,16,3][(tick(3)+i)%4],148+i*12,186,7,5)} //
  TVs en vidriera
  R(9,150,114,1,16);R(9,176,114,1,16);const ad=Math.floor(mod(T,LOOP)/32)%3;
  R([12,15,17][ad],146,100,36,14);txt(["FIZZ-O!","ZEPHYR V8","LUX
  TV"][ad],148,103,[15,12,0][ad]);
  if(N>.4)E(7,146,114,36,1);
  R(28,188,140,26,60);R(0,186,168,30,9);txt("STAR VOYAGE",186,169,N>.3?7:18); //
  cine
  for(let i=0;i<15;i++){const
  on=(i+tick(8))%3==0;if(N>.3&&on){E(7,186+i*2,167,1,1);E(7,186+i*2,177,1,1)}}

R(9,216,152,24,48);txt("ARCADE",216,153,(N>.3&&tick(4)%7)?16:15);E(N>.3?17:1,224,186
,7,14); // videojuegos
  crane(64,105,24,40,96,90,194,84,uOf(4,32),9);
  // calle y vereda
  R(11,0,200,240,7);R(8,0,207,240,18);for(let
x=0;x<240;x+=12)R(12,x,215,6,1);R(11,0,225,240,11);R(10,0,236,240,4);
  for(let
i=0;i<14;i++)R([16,18,17,20,15,28][i%6],hash(i,40)*230,236+(i%2),5+(hash(i,41)*8|0),2); //
  graffiti en el murete

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[30,110,190].forEach(x=>lamp(x,false));
R(28,24,201,9,5);R(25,26,198,2,2);R(9,25,197,4,1);R(23,34,202,4,4);    // persona sin
techo, sentada con sus cosas
R(21,20,222,16,12);R(27,19,220,18,2);txt("NEWS",21,221,12);    // puesto de
diarios
R(11,176,227,12,5);R(8,177,232,2,2);R(8,185,232,2,2);for(let
i=0;i<4;i++)R(i%2?12:15,176+i*3,219,3,2); // carrito de panchos
P8(172,234,12,8,25,tick(2),{hat:12});
R(9,118,217,6,1);u=uOf(30,20);const sn=u>=0?8:3;    // alcantarilla con
vapor
for(let i=0;i<sn;i++){const
a=mod(tick(8)+i*5,24);R(i%2?11:12,120+Math.round(osc(24,i)*2),215-a,2+(a>3),2)}
const nb=Math.floor(mod(T,LOOP)/89*5)%5;for(let i=0;i<nb;i++)R(i%2?8:0,90+i*4,229,4,5);
// basura que se acumula
for(let i=0;i<5;i++)car(cyc(6,280,i*56)-20,214,[18,15,18,22,12][i],1,i%2==0);    //
embotellamiento
for(let i=0;i<6;i++)car(250-cyc(2,280,i*46),223,[18,9,14,18,28,12][i],-1,i%3==0);
crowd(CL,0);
// --- 16 micro-eventos ---
u=uOf(2,10);if(u>=0){const
tx=walkIn(136,u,-30,260);R(12,tx,203,16,9);txt("DAYE",tx+1,204,15);R(21,tx+16,206,6,6);R(8
,tx+2,212,3,2);R(8,tx+17,212,3,2)} // 1 reparto
u=uOf(8,8);if(u>=0){const
tx=u<.4?lerp(250,104,u/.4):u<.7?104:lerp(104,-30,(u-.7)/.3);car(tx,223,18,-1,1);if(u<.55)P8(u
<.15?lerp(70,100,u/.15):100,234,15,27,25,tick(6),{arm:u>.15,hair:13})} // 2 taxi
u=uOf(14,6);if(u>=0){const
x=lerp(-10,250,u);R(8,x,231,3,3);R(8,x+7,231,3,3);R(9,x+2,229,7,1);P8(x+3,230,17,8,26,0,{b
ag:29})} // 3 mensajero
u=uOf(20,14);if(u>=0)P8(walkIn(190,u,250,250),234,20,21,26,tick(6),{arm:u>.4&u<.6}); //
4 compra panchos

u=uOf(26,12);if(u>=0){P8(walkIn(150,u,240,100),206,27,8,25,tick(6),{bag:13});P8(walkIn(162
,u,250,120),206,15,8,26,tick(6)+1,{})} // 5 vidriera
// 6 vapor (30 s) arriba
u=uOf(36,12);if(u>=0)for(let
i=0;i<5;i++)P8(lerp(-40,260,u)-i*9,234,[15,20,12,28,14][i],[21,8,23,21,8][i],i%2?26:25,tick(6)+i
,{sign:i%2==0}); // 7 marcha desarme
u=uOf(44,12);if(u>=0)for(let
i=0;i<3;i++)P8(walkIn(74+i*10+Math.round(osc(24,i)*3),u,-20,-20),206,[29,21,9][i],27,i==1?26
:25,tick(6)+i,{sign:1,hat:i==0?18:null}); // 8 piquete

u=uOf(48,14);if(u>=0){P8(walkIn(38,u,120,120),206,27,8,25,tick(6),{bag:13});if(u>.35&&u<.8
5)R(12,u<.55?37:33,200,1,2)} // 9 café para la persona sin techo

u=uOf(50,12);if(u>=0)P8(lerp(250,-10,u),234-(tick(4)&1),16,21,26,tick(6),{phones:1,hair:31});
// 10 walkman
// 11 persiana (52 s) arriba

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u=uOf(58,8);if(u>=0){P8(walkIn(40,u,-20,-20),234,9,8,25,tick(6),{});if(u>.4&&u<.85)R(12,44,2
26,3,3)} // 12 canillita
u=uOf(63,3);if(u>=0&&tick(8)%2){E(30,145,99,38,1);E(30,145,114,38,1)} // 13 cambio de
anuncio
u=uOf(72,16);if(u>=0&&u<.8){const
x=lerp(250,199,Math.min(1,u*1.4));P8(x,206,28,8,25,tick(6),{});P8(x+5,206,15,27,26,tick(6)+
1,{hair:13})} // 14 cine
u=uOf(74,14);if(u>=0)for(let i=0;i<3;i++){const
x=lerp(120,228,u)-i*8;if(x<226)P5(x,206,[17,15,18][i],tick(6)+i,0)} // 15 chicos al arcade
u=uOf(86,9);if(u>=0){const
x=u<.3?lerp(-30,84,u/.3):u<.7?84:lerp(84,260,(u-.7)/.3);R(19,x,205,20,9);R(12,x+20,207,6,7);
R(8,x+3,213,3,2);R(8,x+19,213,3,2)} // 16 basura
}
// ===== CIUDAD SOCIALISTA (x 240..479) =====
function drawRight(){let u;
R(19,240,178,240,22);
R(9,318,40,2,130);R(10,315,70,8,6);R(9,316,30,4,10);if(N>.4)E(15,317,28,2,2); //
torre de TV
R(9,420,140,60,60); // fábrica colectiva
const
pol=uOf(64,10)>=0;[428,446,466].forEach(cx=>{R(13,cx,92,5,48);R(12,cx,96,5,2);R(12,cx,10
6,5,2);puffs(cx,92,pol?10:6,pol)});
R(15,422,144,56,7);txt("СЛАВА ТРУДУ!",422,144,12);
R(24,420,186,60,14);for(let y=188;y<200;y+=3)R(23,420,y,60,1); // campo
del koljós
u=uOf(0,90);if(u>=0){const
x=lerp(478,420,u);R(15,x,184,14,8);R(4,x+2,181,5,3);R(8,x+2,192,4,3);R(8,x+9,192,4,3)} //
cosechadora
R(11,242,90,58,110);for(let y=90;y<176;y+=11){R(10,242,y,58,1); //
brezhnevka
for(let x=244;x<298;x+=8){if(hash(x,y)<N*.8)E(7,x+1,y+3,4,4);else R(3,x+1,y+3,4,4);
if(hash(x+1,y)<.25)R([16,20,15,18][(x+y)%4],x,y+8,6,2))} // balcones con
ropa y flores
R(12,242,178,58,8);txt("ФАКТОХОМ",244,178,15);R(1,244,187,54,13);R(9,272,188,8,12);
R(15,291,190,6,15);R(12,292,192,4,3); // máquina de soda
R(12,302,150,38,50);for(let
y=150;y<200;y+=12)R(10,302,y,38,1);winGrid(302,150,38,48,5,6,9,71); // jruschovka
R(10,302,142,38,8);crane(300,200,44,288,346,334,194,132,uOf(4,32),10);
R(10,342,60,28,140);R(21,344,64,24,44); // mosaico espacial
original
for(let y=64;y<108;y+=2)for(let x=344;x<368;x+=2)if(hash(x,y)<.3)R(17,x,y,1,1);

R(18,346,67,7,7);R(12,359,70,4,16);R(15,359,67,4,3);R(29,360,86,2,2);winGrid(342,112,28,
74,4,5,9,61);
const dk=uOf(70,16)>=0||N>.4;R(12,372,148,48,52);R(11,370,144,52,4);txt("ДОМ
КУЛЬТУРЫ",371,150,15); // Casa de la Cultura
for(let i=0;i<5;i++){if(dk)E(7,379+i*8,162,4,10);else R(1,379+i*8,162,4,10)}

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    for(let r=0;r<5;r++){const w=osc(40,r*.6)>0?1:0;R(15,379,126+r,8+w,1);R(9,378,126,1,20)}
// bandera
    R(21,420,190,20,16);txt("ПЕЧАТЬ",420,191,12); // quiosco
    // calle, tranvía y plaza

R(11,240,200,240,7);R(9,240,207,240,18);R(8,240,213,240,1);R(8,240,217,240,1);R(8,240,1
86,240,1);
    [250,330,410].forEach(x=>R(9,x,184,1,24));
    R(12,240,225,240,11);R(10,240,236,240,4);for(let
x=240;x<480;x+=8)R(x%16?15:18,x,237,4,2); // murete pintado

[250,440].forEach(x=>{R(19,x,227,14,4);R(16,x+2,227,1,1);R(15,x+6,228,1,1);R(18,x+10,22
7,1,1)}); // canteros
    R(15,400,226,1,8);R(15,410,226,1,8);R(15,400,226,11,1);const
sw=Math.round(osc(30)*2);R(9,405+sw,227,1,4);R(18,404+sw,231,3,1); // hamaca pintada
    R(9,300,212,1,22);R(10,301,212,3,2); // altoparlante
    R(18,326,226,14,7);txt("KBAC",327,226,15);R(8,327,233,2,2);R(8,337,233,2,2); //
barril de kvas

R(23,366,229,8,2);R(12,366,228,8,1);P8(360,234,23,8,25,0,{hair:11});P8(376,234,9,8,26,0,{
hair:11}); // ajedrez
    [262,342,412].forEach(x=>lamp(x,true));
    for(let k=0;k<2;k++){const x=cyc(2,320,k*160)-40;R(15,x,204,40,8);R(12,x,209,40,3); //
tranvía
        for(let i=0;i<6;i++){if(N>.4)E(7,x+3+i*6,205,4,3);else
R(4,x+3+i*6,205,4,3)}R(9,x+18,190,1,14)}
        {const
x=500-cyc(3,300,50);R(12,x,213,34,9);R(21,x,217,34,2);R(9,x+8,187,1,26);R(9,x+10,187,1,2
6); // trolebús
            if(N>.5&&tick(10)%17==0)E(7,x+8,186,2,1)}
            crowd(CR,240);
            // --- 16 micro-eventos ---
            u=uOf(2,8);if(u>=0){for(let i=0;i<3;i++){const
a=mod(tick(6)+i*4,12);R(12,305+a,211-a/2,1,2)}P8(288,234,21,21,25,0,{arm:tick(2)%2})} // 1
altoparlante y gimnasia
            // 2 grúa de paneles (4 s) arriba
            u=uOf(6,12);if(u>=0){const x0=lerp(236,490,u);P8(x0,234,12,21,25,tick(6),{hat:15});for(let
i=0;i<5;i++)P5(x0-6-i*5,234,[15,18,17,20,16][i],tick(6)+i,0)} // 3 jardín de infantes
            u=uOf(12,6);if(u>=0){const
a=u>.3&&u<.8;P8(walkIn(286,u,240,240),206,21,21,25,a?0:tick(6),{arm:a})} // 4 máquina de
soda
            u=uOf(18,16);if(u>=0)for(let
i=0;i<3;i++)P8(walkIn(320-i*6,u,240,240),234,[23,15,21][i],[8,9,27][i],25,tick(6)+i,{hat:i==1?16
:null}); // 5 kvas
            u=uOf(22,10);if(u>=0){P8(452,199,16,23,25,0,{hat:16});if(tick(8)%2)E(30,450,193,3,2)} // 6
obrera soldadora

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    u=uOf(28,22);if(u>=0){for(let
i=0;i<6;i++)P8(walkIn(268-i*5,u,236,276),206,[23,15,21,28,12,20][i],[8,9,27,8,21,9][i],i%3?25:
26,0,{hat:i%2?16:null}); // 7 cola
    if(u>.5){const
x=u<.6?lerp(480,250,(u-.5)/.1):u>.85?lerp(250,480,(u-.85)/.15):250;R(12,x,203,16,9);txt("XΠ
ΕΒ",x+1,203,15))} // 8 llega el pan
    u=uOf(34,8);if(u>=0)car(lerp(480,236,u),214,20,-1,0); // 9 el único auto
particular
    u=uOf(38,12);if(u>=0)for(let i=0;i<5;i++)P5(lerp(250,480,u)-i*6,234,21,tick(6)+i,1); // 10
pioneros
    u=uOf(44,8);if(u>=0&&tick(2)%2)R(31,370,227,1,1); // 11 jugada de
ajedrez
    u=uOf(52,10);if(u>=0)for(let
i=0;i<4;i++)P8(lerp(420,300,u)+i*7,206,[29,21,23,9][i],27,25,tick(6)+i,{hat:i%2?9:null}); // 12
cambio de turno
    u=uOf(58,10);if(u>=0){P8(430,206,9,8,25,0,{hat:9});if(tick(4)%2)R(12,434,199,2,2)} //
13 hombre de gris que anota

u=uOf(62,8);if(u>=0){P8(walkIn(356,u,480,480),206,28,21,26,tick(6),{});if(u>.3&&u<.7)R(21,3
60,200,2,3)} // 14 vaqueros bajo el sobretodo
// 15 humo denso (64 s) arriba; 16 Casa de la Cultura (70 s)
    u=uOf(70,16);if(u>=0)for(let i=0;i<4;i++){const
x=lerp(480,396,Math.min(1,u*1.5))+i*6;P8(x,206,[16,12,15,28][i],8,i%2?26:25,tick(6)+i,{})}
    u=uOf(88,7);if(u>=0)for(let
i=0;i<2;i++)P8(lerp(250,480,u)+i*20,234,29,21,25,tick(6)+i,{hat:24}); // barrenderas
}
// ===== GUÍA (8 etiquetas por lado) =====
const LAB=[[21,46,"Torre de estilo internacional"],[88,112,"Banco: cotizaciones en
vivo"],[164,98,"Carteles luminosos"],[53,178,"Negocio que cierra (desempleo 10,8 %)"],
[29,196,"Persona sin techo, con sus cosas"],[200,138,"Cine y salón de
videojuegos"],[160,210,"Taxis y embotellamiento"],[120,236,"Grafiti: arte y conflicto"],
[270,88,"Brezhnevka de 9 pisos"],[320,140,"Jruschovka de paneles"],[356,60,"Mosaico
espacial"],[270,176,"Gastronom: colas y escasez"],
[396,144,"Casa de la Cultura"],[450,90,"Fábrica: empleo y
contaminación"],[294,210,"Altoparlante en el poste"],[370,224,"Ajedrez en la plaza"]];
let guide=false,S=1;const ov=document.getElementById('ov');
function renderGuide(){ov.innerHTML="";if(!guide)return;LAB.forEach(l=>{const
d=document.createElement('div');d.className='lb';
    d.style.left=l[0]*S+'px';d.style.top=l[1]*S+'px';d.textContent=l[2];ov.appendChild(d)}})
document.getElementById('bg').onclick=e=>{e.stopPropagation();guide=!guide;renderGuide(
)};
document.getElementById('stage').onclick=(e=>{guide=!guide;renderGuide()});
// ===== SONIDO PROCEDURAL (arranca solo después de un toque) =====
let AC=null,gL,gR;
document.getElementById('bs').onclick=e=>{e.stopPropagation();
    if(AC){AC.state==='running'?AC.suspend():AC.resume();return}
    AC=new (window.AudioContext||window.webkitAudioContext)();

```

```

const
len=AC.sampleRate*2,nb=AC.createBuffer(1,len,AC.sampleRate),d=nb.getChannelData(0);
let s=12345;for(let i=0;i<len;i++){s=(s*1103515245+12345)&0x7fffffff;d[i]=(s/0x7fffffff)*2-1}
function bed(pan,freq){const src=AC.createBufferSource();src.buffer=nb;src.loop=true;
const f=AC.createBiquadFilter();f.type='lowpass';f.frequency.value=freq;const
gn=AC.createGain();gn.gain.value=0;
const
p=AC.createStereoPanner?AC.createStereoPanner():null;src.connect(f);f.connect(gn);
if(p){p.pan.value=pan;gn.connect(p);p.connect(AC.destination)}else
gn.connect(AC.destination);src.start();return gn}
gL=bed(-.7,700);gR=bed(.7,350));
function audioTick(){if(!AC)return;const
day=1-N;gL.gain.setTargetAtTime(.05+.05*day,AC.currentTime,.3);gR.gain.setTargetAtTime(
.04+.03*day,AC.currentTime,.3)}
// ===== ESCALADO ENTERO Y BUCLE =====
function fit(){const
dpr=window.devicePixelRatio||1;S=Math.max(1,Math.floor(Math.min(innerWidth/W,innerHeight/H)));
if(innerHeight>innerWidth)S=Math.max(1,Math.floor(innerHeight*.9/H)); // vertical: scroll
horizontal
cv.width=W*S;cv.height=H*S;cv.style.width=W*S+'px';cv.style.height=H*S+'px';

document.getElementById('stage').style.width=W*S+'px';dc.imageSmoothingEnabled=false;renderGuide()}
addEventListener('resize',fit);fit();
const t0=performance.now();
function
frame(now){T=mod(START+(now-t0)/1000*SPEED,LOOP);P=T/LOOP;N=nightOf(P);
g.clearRect(0,0,W,H);drawSky();drawCelestial();drawLeft();drawRight();
R(8,239,0,2,200); // costura entre ciudades
birds();drawRiver();audioTick();

dc.imageSmoothingEnabled=false;dc.drawImage(buf,0,0,W*S,H*S);requestAnimationFrame
(frame)}
requestAnimationFrame(frame);
</script></body></html>

```